Antiocclusion Infrared Aerial Target Recognition With Vision-Inspired Dual-Stream Graph Network

Xi Yang, *Member, IEEE*, Shaoyi Li, *Member, IEEE*, Liang Zhang, Binbin Yan, *Member, IEEE*, and Zhongjie Meng, *Member, IEEE*

***Abstract*— Infrared-guided aircraft usually faces complex interference, such as natural backgrounds and infrared decoys, which decrease the performance of infrared target recognition models. A great challenge is the strong interference posed by the combination of target maneuvering and the dense and continuous deployment of infrared decoys, severely limiting the target recognition and anti-interference capabilities of infrared imaging seekers. To address the issue of target occlusion in the imaging phase, this article proposes an antiocclusion recognition model. First, inspired by the ventral pathway's capabilities of shape information processing, a skeleton-based shape feature description model is proposed based on target keypoints, which incorporates the prior knowledge about the spatial relationships of skeleton nodes inherent in the shape features of the target. Second, inspired by the dorsal pathway's capabilities of motion information processing, a motion-based feature description model is proposed using optical flow, which improves keypoint estimation performance by integrating motion information. Moreover, considering the influence of local keypoint occlusion on the stability of the target's skeleton shape representation, a learning module is introduced, which transfers high-order semantic information between nodes to strengthen the connections between keypoints while suppressing the transmission of meaningless features. Finally, an antiocclusion recognition model is proposed based on a vision-inspired dual-stream graph network (VIDSGN) for environments with interference and occlusion, which can estimate the positions of occluded nodes using known keypoints and high-order information. The experimental results on the infrared dataset constructed in this article indicate that our model has stable representation capabilities for target shape structures in the presence of partial occlusion.**

***Index Terms*— Aerial object recognition, antiocclusion, graph network, visual pathway inspiration.**

I. Introduction

IN COMPLEX aerial combat environments, in the imaging phase of infrared-guided aircraft attacks, the deployment of infrared decoys by airborne targets causes partial occlusion of the targets, resulting in the loss of target feature information and severely interfering with the stable recognition and tracking of the infrared seekers. As shown in Fig. 1, when facing interference and occlusion, traditional target recognition models based on multifeature fusion and template matching fail to extract local structural information from the unoccluded parts. This easily contaminates the feature information of the complete target template, leading to template degradation and incorrect target recognition [1].



Fig. 1. Complex interference countermeasure environment in the air.

Target recognition models constructed based on convolutional neural networks (CNNs) can effectively identify the unoccluded parts of the targets, but they cannot effectively estimate the overall target position using the information from the unoccluded parts and fail to estimate global structure from local features [2]. In the imaging phase of aerial combat scenarios, there are various challenges, such as the target's high-speed maneuvers, significant feature changes caused by long-term partial occlusion accompanied by infrared decoys, local feature information loss, and drift of tracking points. These complex interference and occlusion cannot be captured without considering the temporal dependencies between video frames. Meanwhile, CNN models have difficulties in recognizing airborne targets with heavily occluded interference. Most advanced target recognition models estimate the target's position in each frame and then perform data association and refinement [1]. Although such models have made some

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progress, they are prone to failure in highly occluded interference scenarios because they heavily rely on spatial visual features that are missing in occlusion conditions. Thus, the key to extracting infrared target features in the air lies in adopting stable and effective approaches to describe target features, which requires extracting local structural features with good discriminative ability to recognize the unoccluded parts and improve the model's anti-interference performance.

Studies have shown that the human visual system employs two visual pathways in the target recognition process: the ventral pathway for object recognition and the dorsal pathway for extracting motion components in the visual field. The ventral pathway plays a key role in processing the shape information of targets [3], [4], [5]. Meanwhile, the visual neural pathway can exploit partial structural information of known objects to predict the overall shape and position and utilize local shape information to reconstruct the global shape structure [3], [4]. Skeletal representations are crucial for context-aware recognition tasks, as they succinctly characterize the overall topological structure of the target object and facilitate computing visual similarities between different skeletons [6]. Also, the processing and analysis of object motion information are significant for visual target recognition in the brain. This function is primarily conducted by the dorsal information pathway in the visual cortex. This pathway extracts the motion of all objects in the visual field, and by using a temporal δ, the visual attention system selects the target motion components from the background [7]. In this pathway, the processing of visual motion information can be divided into two levels: low-level processing and high-level processing [7]. Specifically, in low-level processing, local optical flow is computed to extract motion information; in high-level processing, advanced scene representations encoding object dynamics, including motion trajectories, shape models, and mechanical properties, are derived from the optical flow field [8]. Therefore, optical flow features provide an important approach to visual motion analysis of targets, and they play a crucial role in the dorsal pathway's processing of target motion information [9], [10]. In summary, the information processing functional characteristics and performance of the ventral pathway and dorsal pathway in target recognition have certain advantages for target recognition algorithm development and engineering applications.

To handle the partial occlusion of targets, inspired by the information processing characteristics of the ventral and dorsal visual pathways, we introduce skeleton features and optical flow features for describing airborne infrared targets. Then, an antiocclusion recognition model is constructed based on a vision-inspired dual-stream graph network (VIDSGN) to estimate the target's skeleton structure and position in the case of partial occlusion, thus achieving robust target recognition in the presence of interference. The main part of the adopted antiocclusion strategy is building a dual-stream graph convolutional network (GCN), and the entire recognition process involves extracting spatial and temporal features of the target region and then utilizing graph networks to estimate the positions of the occluded regions, thereby recognizing the complete target. This approach aims to address the issue of unstable model recognition in the presence of partial occlusion and improve the model's accuracy and robustness. The core of the model proposed in this article is the graph reasoning module that models and analyzes the spatial structural relationships between key parts of the target. By describing the spatial and motion features of each key part, a skeleton graph model connecting the features of each part is constructed, and relational reasoning is carried out on the graph.

Specifically, the proposed model first constructs a skeleton graph model to represent the spatial relationships between various parts. It constructs the skeleton graph module by using the recognizable target region, where key parts in the image are represented as graph nodes and the spatial relationships between nodes are encoded with graph edges. Second, a node optical flow model is employed to represent the temporal states of each part. The spatiotemporal features are fused to capture the relationships between features within the corresponding nodes, specifically designed for message passing in graph inference. Then, the occluded parts are encoded with position information by the recognition model, and the graph inference module is adjusted to reconstruct the complete shape of the occluded target. In this way, this recognition process simulates the ventral and dorsal pathways' process of visual shape information from lower to higher levels. Finally, by constructing the graph inference module for the scenario of partial occlusion, the positions of occluded skeleton nodes are predicted based on known node features. The main contributions of this article are summarized as follows.

1) A novel framework is proposed for antiocclusion recognition of airborne targets, which utilizes graph models and graph convolution-based inference modules to infer the semantic structure of target key nodes. The spatial and temporal semantic relationships between nodes are assigned to occluded nodes to achieve high robustness against interference.
2) Inspired by the ventral visual pathway, a target skeleton graph model is introduced where a set of target key parts in the coordinate space is treated as nodes and projected into a skeleton graph model that can efficiently compute spatial relationships. Then, the positions of occluded nodes are estimated through reasoning.
3) Inspired by the dorsal visual pathway, a node optical flow model is proposed, which calculates the motion states of target key nodes using an optical flow model. By incorporating the optical flow model with the skeleton features, the spatiotemporal features of the target are described more comprehensively.

II. Related Work

A. Aerial Target Recognition

With deep neural network technology being adopted as the mainstream for designing target recognition models and its demonstrated ability to learn complex features, researchers have shown great interest in the latest advancements in the field of airborne infrared target recognition and have attempted to improve and apply them to anti-interference recognition of airborne targets. Object detection models such as SSD

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[11] and YOLO [12] have been used as data-driven end-to-end approaches for airborne target recognition, and they have achieved promising results in terms of higher recognition performance and improved target classification capabilities, thus greatly advancing the level of intelligent recognition of airborne infrared targets. However, in increasingly complex aerial battlefield environments, numerous challenges still need to be addressed. Li et al. [13] presented a method that combines traditional features with Bayesian networks for anti-interference recognition of infrared-guided aircraft. This method demonstrates good performance in target and interference classification problems, addressing the issue of target occlusion to some extent. Yang et al. [14] developed an inference-based recognition model combining convolutional features with Bayesian networks. Specifically, a CNN architecture was utilized to extract descriptive features from infrared imagery. A 2DPCANet filter was then applied to reduce dimensionality and select salient details within candidate regions. Finally, a Bayesian network classifier was constructed to perform probabilistic reasoning for infrared target recognition. Zhang et al. [1] constructed an anti-interference recognition model for airborne infrared targets based on the deep CNN DNET, which introduces feature attention mechanisms and multiscale dense connection modules to improve the capability of recognizing airborne targets under scale variations and interference occlusion. MF-GCN [2] introduces a GCN-based machine vision model for aerial target recognition. The model advances keypoint estimation by predicting optical flow information, and an adaptive graph fusion module is designed to generalize regular network features to the target skeleton graph model. MSGSM [15] proposed a deep learning model based on a GCN for anti-interference recognition of aerial targets. The model establishes a target skeleton graph model and learns high-order relationships and topological information to infer occluded key points and realize target recognition. Hu et al. [16] proposed a novel model based on the Siamese network for anti-interference target recognition. By exploiting traditional features such as aspect ratio and average contrast, the model exhibited certain anti-interference capabilities when tested on unseen data.

B. GCN

The GCN has garnered significant interest owing to its ability to model non-Euclidean structured data such as graphs. Based on the formulation of graph convolutional operations, GCN approaches are broadly categorized as either spatial- or spectral-domain methods, with the former preprocessing the data and defining convolutions directly over nodes and their local neighborhoods [17], [18], [19], and the latter exploits the eigenvalues and eigenvectors of the graph Laplacian matrix [20]. Many studies have applied spatial GCN to computer vision tasks; especially, it is widely used in human skeleton modeling [21], [22]. The role of GCN is to capture the situation dependence and intrinsic topological structure characteristics of human behavior and combine them with spatiotemporal information for modeling. In skeletal modeling based on graph models, the graph structure of the human skeleton needs to be constructed first [23]. This can be done by connecting body joint points to form nodes and edges of the graph. Each joint point represents a human body part, and the edges represent the connection relationships between different joint points. Then, the graph convolutional neural network is used to extract and learn features of the graph structure. By applying convolution operations on each node, on the one hand, GCN can use the connection relationships between nodes to propagate and aggregate information [24]. On the other hand, it can capture the structural and dynamic changes of human posture, and iterative calculations can be performed from the input higher level semantic information that is extracted from the features [23], [24]. In our work, drawing on the ideas of GCN for human skeleton modeling, GCN is used for antiocclusion recognition to extract aerial target structure modeling.

III. Vision-Inspired Dual-Stream Graph Network

The human brain can identify potential targets in complex scenes, and it exhibits remarkable recognition ability for moving objects. The underlying neural mechanisms behind this recognition ability are the core of visual perception and cognition research. Extensive studies have demonstrated that the hierarchical structure of the visual neural system, the parallel processing of information, the interaction of feedforward and feedback information, as well as the close integration with the eye movement system enable visual information to be processed accurately and efficiently, which lay a solid foundation for rapid target recognition. The overall structure of information interaction in the visual pathway is illustrated in Fig. 2. The human visual system exhibits a striking ability to rapidly identify targets amidst complex scenes and track moving objects. Elucidating the neural mechanisms underlying this recognition capability is central to visual perception and cognition research. Substantial evidence indicates that the hierarchical organization of visual pathways, parallel processing of visual information, interplay of feedforward and feedback signals, and tight integration with the oculomotor system enable robust, efficient, and reliable analysis of visual inputs [25], providing a critical foundation for rapid target recognition. The overall architecture of information flow in the visual system is depicted in Fig. 2.

The visual system in the brain is composed of two principal processing streams: the ventral pathway, also known as the “what” pathway, and the dorsal pathway, known as the “where” pathway. The ventral visual stream is responsible for analyzing and recognizing visual features including shape, color, and category information. It starts from the primary visual cortex (V1) and passes through areas such as V2 and V4, finally reaching the inferior temporal (IT) cortex [26], [27]. Meanwhile, the dorsal pathway is responsible for processing spatial location information and motion information of objects. It starts from V1 and passes through V2 before reaching the middle temporal area (MT or V5). From there, it proceeds to the posterior parietal (PP) cortex [28], [29]. The prefrontal cortex, a higher order visual area, is critically involved in visual object classification, scene understanding, feature-based attention, and spatial attention [28], [29]. Also,

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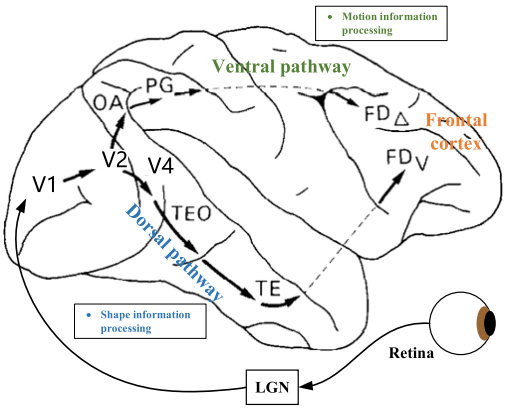


Fig. 2. Schematic of two visual neural pathways.

it exhibits rich connectivity with both the ventral and dorsal pathways.

Both dorsal and ventral visual streams process information in a hierarchical manner. As knowledge of this hierarchical organization has advanced, investigation of the neural substrates underlying visual recognition has progressed in an increasingly detailed and precise fashion. The primary visual cortex (V1) and visual area V2 contain simple and complex cells, which exhibit orientation selectivity as originally discovered by Hubel and Wiesel [30], and they respond maximally to stimuli with a specific orientation within their receptive fields. Complex cells have larger receptive fields and respond maximally to stimuli with a specific orientation. The IT cortex is located at a higher level of visual information processing. Studies have shown that there are both transient and sustained neural activities in V1 and IT during the visual information processing of objects [31]. To sum up, the hierarchical structure of the visual neural pathways involves visual information processing at different levels and stages. The complexity of information processing increases progressively along the visual pathways and culminates in higher level visual areas responsible for processing high-level information. Both pathways demonstrate hierarchical processing from simple to complex and from low-level to high-level feature information.

Based on the visual neural pathways' information processing capability, to specifically overcome the challenges of target representation and recognition due to strong adversarial interference such as target maneuvering and partial/long-term occlusion, this article proposes a vision-inspired dual-stream antiocclusion recognition model that integrates key information processing functions such as shape feature processing and motion information processing. To solve the problem of partially occluded targets and the loss of feature information caused by continuous interference, inspired by the shape information processing capabilities of the ventral pathway, this article constructs a target skeleton module, which extracts the skeleton keypoints of aerial targets using a deep hierarchical network and decomposes and trains them according to the shape structure of the target skeleton. Meanwhile, the proposed model reconstructs the complete target shape based on the unoccluded local shape structure features to improve the stable recognition ability of the model in partially occluded scenarios. Furthermore, inspired by the motion information processing capabilities of the dorsal pathway, this article constructs a target node optical flow module to address the issue of drastic feature changes and local feature information loss caused by long-term partial occlusion of infrared decoys. It uses an optical flow field model to extract target motion features and effectively integrates the target motion and shape features to predict the positions of occluded nodes, thereby enhancing the model's ability to resist partial occlusion. In addition, by combining the temporal and spatial features of the targets, this article constructs a graph inference module to represent the spatial semantic information between the target keypoints. The module combines spatial–temporal information inference based on the multiregion spatial structure and predicts the target position, achieving target prediction and antiocclusion.

A. Graph-Based Skeleton Module

Substantial evidence indicates that the human visual system exhibits selective responses to salient regions within objects of interest [32]. To encode different objects, these prototypical targets are decomposed into multiple regions of interest, and these visually distinct and spatially nonoverlapping regions are referred to as keypoints. Each keypoint serves as a building block in the model. For modeling purposes, keypoints are representations of the target at multiple scales and different appearances in various conditions, forming the basis for constructing the target skeleton structure graph model. Inspired by behavioral and neuroimaging research that highlights the visual system's sensitivity to skeleton structures and increasing evidence that object perception is best described by shape representations centered around the object [33], this study chooses the representation of the target's 3-D skeleton structure in a 2-D image. In this section, a target skeleton structure graph model is proposed, which utilizes skeleton structure features to describe target shape characteristics. This model enables accurate and robust measurement of similarity between the target representation and potential noise or interference [34], [35].

Before constructing the target skeleton model, it is necessary to extract the keypoint features of the visible region of the target. Aligning with the human visual system's information processing and detection process for visual targets, keypoint extraction is performed for each frame in a bottom-up object detection process [36], [37], [38]. In each frame, each keypoint is detected, cropped, and scaled to a fixed size. Taking the scaled image as input, the keypoint feature extraction backbone network outputs two parts: keypoint heatmaps and per-pixel keypoint regression results. The keypoint heatmaps $\mathbf{H}$ consist of K maps $\{\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_k\}$, with the number of keypoints $K = 9$ set in this dataset. Through the heat value at that position, each keypoint heatmap implies the degree of presence of the keypoint at each position. The per-pixel keypoint regression results generate the heatmap $\mathbf{H}$, which indicates the degree to which each position is the center of a pose. Let $\mathbf{H}_{ijk}$ be the probability value at position (i, j) of the kth heatmap. The position of the kth keypoint is computed

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as follows:

$$p_k^* = \arg\max_{(i,j)} \mathbf{H}_{ijk}. \tag{1}$$

Based on the center and scale information of the cropped image, the predicted keypoint location p_k^* in the heatmap can be transformed into image coordinates. The training loss for the keypoint detection network is calculated based on the heatmaps. The cropped target image is resized to a fixed dimension, with the corresponding annotations appropriately transformed to heatmap coordinates. Let p_k^* be the ground-truth position of the kth node in the heatmap. The standard heat map is generated based on a 2-D Gaussian distribution model. In the first stage of training, the keypoint detection network is trained by minimizing the following loss function, where H and W, respectively, represent the height and width of the heat map and K is the number of keypoints:

$$L^{s1} = \sum_{i}^{H} \sum_{j}^{W} \sum_{k}^{K} \left\| \mathbf{H}_{ijk}^{dt} - \mathbf{H}_{ijk}^{gt} \right\|_2^2 \tag{2}$$

where $\mathbf{H}^{dt}$ represents the heatmap of keypoints detected by the model and $\mathbf{H}^{gt}$ represents the ground truth of the keypoints. The nodes in the constructed skeleton graph model are represented by heat maps of the associated feature space [39], [40]. In this section, an intuitive graph $G(V, E)$ is constructed, where the vertex set $V = (V_i | i = 1, 2, \ldots, K)$ is set to K nodes. The physical connections between each pair of keypoints in the skeleton graph are encoded as the edge set $E_{ij} = \{V_i, V_j\}$. The adjacency matrix $\mathbf{A} = \{a_{ij}\}$ of G represents the connectivity between nodes, and a_{ij} is represented as follows:

$$a_{ij} = \begin{cases} 1, & V_i \text{ and } V_j \text{ are neighbors or } i = j \\ 0, & \text{others.} \end{cases} \tag{3}$$

Moreover, the visual nervous system not only perceives the shape keypoints but also preserves stable spatial relationships among the nodes. These relationships remain invariant under different viewpoints, and they are recognized and labeled in the visual neural pathway based on their relatively stable spatial positions [20]. The target is identified to belong to a specific category only when the structural information is sufficiently close to the information of the target prototype. For infrared aerial target recognition, it is important to detect and index the structural relationships among the target parts and compare them with prototypes to determine the target's category and position. To encode such spatial relationships, we propose a graph-based skeleton module where the nodes of the graph correspond to the target keypoints, and the edges encode relative distance and proportionality constraints between them. Each pair of keypoints V_i and V_j is connected by an edge parameter e_{ij}, representing the relative change constraint that is necessary to maintain the integrity of the entire target.

As shown in Fig. 3, the direct interactions between the node set V and their interdependencies E form the skeleton graph module $G(V, E)$ and the complexity of the module corresponds to its sparsity. Sparsity carries physical meaning: for most objects, the component layout and resulting viewpoints are not densely statistically dependent, and a sparse interaction graph sufficiently constrains target deformations. Enforcing sparsity during learning poses an optimization problem amenable to efficient solutions.

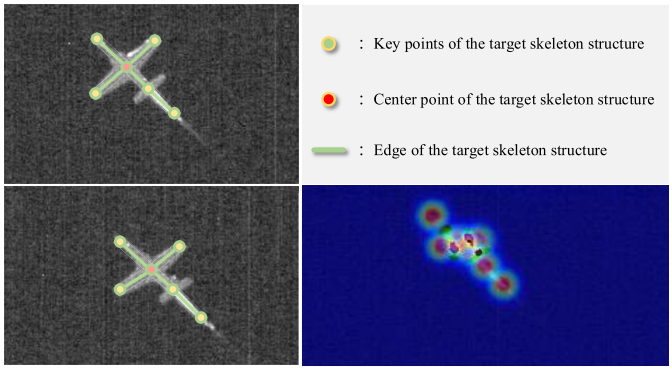


Fig. 3. Schematic of the node feature construction method of the skeleton graph module.

Based on the preceding analysis, the skeleton graph module comprises a set of variable nodes with spatial structural constraints encoded by an undirected graph. In particular, each node is independent of other variables conditioned on its neighbors. This graph-based skeleton graph module functions as a spatial correlation module capturing relationships between local target parts. Next, we detail the construction of this spatial correlation module using GCNs.

The GCN aims to efficiently learn the propagation operation that can be directly applied to graph-structured data. According to the GCN [41], given an undirected graph $G(V, E)$, the graph consists of K nodes V and edges E connecting the nodes. The connectivity between nodes represented by the edges is reflected in the adjacency matrix $\mathbf{A} \in \mathbb{R}^{K \times K}$ and the degree matrix $\Lambda_{ii} = \sum_j \mathbf{A}_{ij}$ of the graph. The linear transformation of graph convolution is represented as the product of the graph signal $\mathbf{X} \in \mathbb{R}^{D \times K}$ and the filter $\mathbf{W} \in \mathbb{R}^{D \times C}$, where the column vector $\mathbf{X}_{i\cdot} \in \mathbb{R}^D$ is the feature representation of the ith node

$$\mathbf{V}^l = \hat{\Lambda}^{-\frac{1}{2}} \hat{A} \hat{\Lambda}^{-\frac{1}{2}} \mathbf{X}^\top \mathbf{W} \tag{4}$$

where $\hat{A} = \mathbf{A} + \mathbf{I}$, $\mathbf{I}$ represents the identity matrix, and $\hat{\Lambda}_{ii} = \sum_j \hat{A}_{ij}$. The network outputs the matrix $\mathbf{V} \in \mathbb{R}^{C \times K}$. It is noteworthy that GCN can be stacked with multiple graph convolutional layers, in the form of (5), where each layer is followed by a nonlinear operation, e.g., an ReLU layer. The spatial relationships and constraints between nodes are captured based on the obtained local features and attributes of the target, and the node features are input into the graph convolutional module to represent the relational features between skeleton graph nodes

$$\mathbf{V}^S = \phi_g(\mathbf{V}^l) = g_1(G \otimes \mathbf{V}^l) + g_2(\mathbf{V}^l) \tag{5}$$

where $\mathbf{V}^l$ denotes the local node features, $\mathbf{V}^S$ represents the relational features of the output nodes, and g_1 and g_2 are fully connected layers. The edges in the spatial correlation module connect nodes within the same skeleton, enabling the GCN to model the relative motion and spatial layout of target keypoints within a frame. To capture variability, the edges are modeled as

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connections with flexible lengths and rigidity between nodes. Since not all keypoints are directly linked within a physical object, the model naturally incorporates sparsity, introducing spatial constraints only between key point pairs critical for preserving overall object integrity, especially physically connected points. Consequently, the adjacency matrix of the corresponding skeleton graph is sparse.

B. Node Optical-Flow Module

This section utilizes the motion variation information between target frames to further enhance the antiocclusion recognition performance and suppress the drift of tracking points. By learning and modeling the description of target motion features, the obscured keypoint positions can be predicted in combination with the skeleton features. Following this research idea, this section introduces the node optical flow module on top of the skeleton graph module, which enhances the feature representation capability of the targets by integrating shape and motion feature information. The proposed node optical flow module can well describe the motion variation between target frames, enabling effective differentiation between target and interfering motion fields, even under partial occlusion. Meanwhile, the fusion of skeleton structure features and optical flow features can effectively combine spatial and temporal information of the targets to predict keypoint positions, thereby further improving the model's recognition and antiocclusion performance.

Due to occlusion introduced by video frames, simply applying a single-frame image target recognition model to a sequence of air combat target recognition images may result in missed detections and false detections. Usually, it is necessary to leverage temporal motion information to generate more robust detection [42], [43]. Therefore, this section proposes to use the motion information in the interframe optical flow to predict the target key point position of the current frame from the adjacent frames. As shown in Fig. 4, given the target instance and key point coordinate set J_i^{n-1} of the previous frame I^{n-1} and the optical flow field $F_{n-1\to n}$ of two adjacent frames I^{n-1} and I^n, the corresponding node set coordinate $\hat{J}_i^n$ of the current frame I^n is estimated by back-propagating the node set J_i^{n-1} according to the optical flow field $F_{n-1\to n}$. More specifically, for each skeletal node position (x, y) in J_i^{n-1}, the propagated joint position is $(x+\delta x, y+\delta y)$, where δx and δy are motion displacements at node position (x, y). Then, the node coordinate set $\hat{J}_i^n$ back-propagated boundary is predicted, and the boundary box is expanded as a target recognition candidate box.

After extracting the motion and skeleton features, it is necessary to project these features from the standard coordinate space to the graph domain. This projection decouples positional information from the original grid-based features, yielding a more compact and descriptive node feature representation. The mapping node V in the skeleton model is divided into two groups: the skeleton node $\mathbf{V}^c = \{V_1^c, \ldots, V_K^c\}$, which contains the features of target shape and region information, and the motion node $\mathbf{V}^m = \{V_1^m, \ldots, V_K^m\}$, which stores the motion features that depend on interframe image matching.

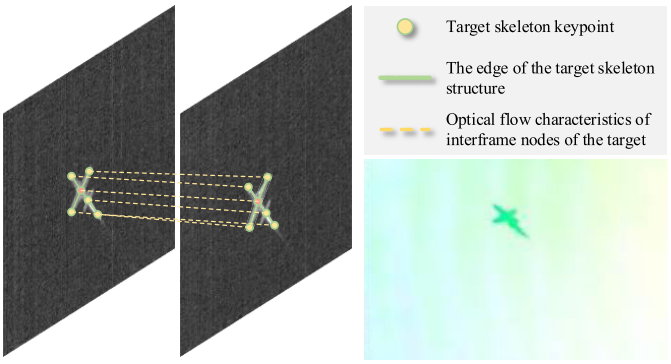


Fig. 4. Schematic of the node optical-flow module.

Specifically, the skeleton feature $\mathbf{V}^c \in \mathbb{R}^{K\times H\times W}$ is obtained from the skeleton graph model, the motion feature $\mathbf{V}^m \in \mathbb{R}^{K\times H\times W}$ is obtained from the optical flow network, and the function $P(\cdot)$ assigns features representing different parts of the target to their corresponding nodes. Let $\mathbf{V}^c \in \mathbb{R}^{C\times K}$ and $\mathbf{V}^m \in \mathbb{R}^{C\times K}$ be the initial node embeddings in the graph space, where C represents the number of channels and K represents the number of nodes. To construct a global graph over a set of regions, $P(\cdot)$ is represented as a linear combination of feature vectors in the grid space, i.e., $\mathbf{F} = P(\mathbf{V})$, and this enables the resulting nodes to accumulate long-range contextual information across the full original feature map

$$F_i = N\left(\sum_{\forall j} \Gamma_{V\to F}\left(V_{ij}\right)\cdot V_j\right) \tag{6}$$

where $N(\cdot)$ is an L2 normalization function along the channel dimension for each node vector and $\Gamma_{V\to F}(\mathbf{V}) \in \mathbb{R}^{N\times K}$ models the projection weights that map features to node vectors. In practice, convolution is applied to $\mathbf{V}^m \in \mathbb{R}^{K\times H\times W}$ twice to obtain a feature map of resolution $K \times H \times W$. Then, a reshaping function is used to generate $\Gamma_{V\to F}(\mathbf{V})$ of resolution $N \times K$, where $N = H \times W$, and K is a hyperparameter independent of the spatial resolution, i.e., the number of keypoints of the target. This results in two types of node features $\mathbf{F}^c = P(\mathbf{V}^c)$ and $\mathbf{F}^m = P(\mathbf{V}^m)$.

C. Feature Fusion Module

We further consider the representational properties of the skeleton and motion nodes. Specifically, motion nodes primarily encode internode correspondence between image pairs while ignoring intraregional relations, whereas skeleton nodes capture discriminative region and shape features. However, two difficulties arise: 1) the representational discrepancy between the feature and motion nodes impedes the effective propagation of informative signals during direct graph reasoning and 2) motion nodes lack constraints on the shape or layout of the optical flow field, preventing extraction of sufficient contextual information for reasoning on a single graph.

To address these challenges, we propose a graph-based feature fusion module (GFM). As depicted in Fig. 5, the GFM decouples skeleton features from the matching process while transferring region and shape priors from keypoint heatmaps

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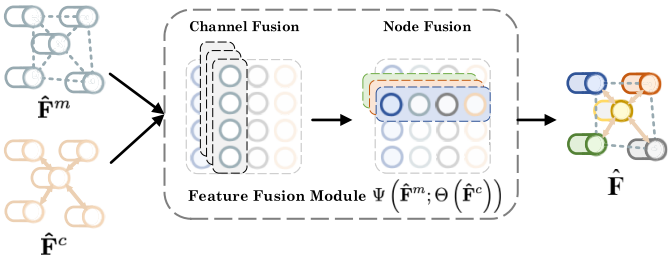


Fig. 5. Schematic of the adaptive feature fusion module.

to motion nodes. The key idea is to leverage discriminative shape and region representations under a global context to guide the learning of motion nodes using adaptive parameters. Inspired by [44], we design an adaptive fusion process between skeleton and motion nodes to predict dynamic parameters that customize motion relation modeling based on image-specific keypoint cues

$$\hat{\mathbf{F}} = \Psi\left(\mathbf{F}^m; \Theta\left(\mathbf{F}^c\right)\right) \tag{7}$$

where $\Theta(\cdot)$ represents the parameter learning module and $\Psi(\cdot)$ denotes the feature fusion module with dynamic weights from features $\Theta(\mathbf{F}^c)$ to motion nodes. This module implements $\Theta(\cdot)$ using a softmax linear projection function. $\mathbf{A}(\cdot)$ is implemented with a two-layer MLP, where the first regular linear function followed by ReLU activation is applied to perform channelwise attention learning, and then, the second linear function $\Theta(\mathbf{F}^c)$ with adaptive kernels exploits node interactions for skeleton-to-motion relation learning. Given skeleton nodes $\mathbf{F}^c \in \mathbb{R}^{C\times K}$, convolutional prediction with channel dimension $(C \times K)$ is used to predict adaptive kernels $\Theta(\mathbf{F}^c) \in \mathbb{R}^{C\times K}$.

D. Graph-Based Reasoning Module

Features for each keypoint are obtained from the skeleton graph module and node optical flow module in the first stage. However, inferior results are produced on occluded joints in this stage. This is mainly because locating obscured target parts from images is highly difficult, as visual shape feature information is lacking with partial occlusion on key target parts. The first-order node features extracted in the first stage do not have the two important properties of aerial targets: exploit different features and processes for accurate localization with priority and consider relationships among keypoints. To compensate for this deficiency, this section introduces a new framework to infer invisible joints from the image background and propose structural cues.

First, human vision performs keypoint localization in two steps [45], [46], [47]. The green keypoints in Fig. 3, including the two wings shown in yellow circles, are roughly localized based on contextual information to determine whether there are wing keypoints in the nearby region. This step can be regarded as a coarse localization process. Second, based on clues from the target structural shape and image context, human vision can still estimate the accurate locations of occluded parts, even under occlusion. Therefore, a design is applied to the target recognition task, where the first stage focuses on the localization process to improve the recall of keypoints, and the second stage focuses on the refinement process to improve prediction and localization accuracy. This section mainly elaborates on the second stage, which performs coordinate refinement based on features corresponding to the keypoints selected in the first stage, to accurately estimate the occluded key parts of the target.

Although the spatial skeleton information and motion information of different keypoint regions is known, target recognition with incomplete target structure and anti-infrared interference occlusion is still difficult. To address this issue, it is necessary to utilize higher order structural semantic features and the GCN methods [48] to model relational information. In GCNs, the spatial skeleton features and motion features of different regions of interest are represented as nodes. Incorporating structural relationships between key target parts into local representations aids in predicting occluded areas. When propagating messages between nodes, both node features and higher order structural semantics encoded in edge features can be leveraged jointly.

However, predicting occluded regions faces one remaining challenge: features of occluded areas are often uninformative. Propagating such noisy features in the graph network introduces additional interference for occluded region prediction. In this case, node interaction information is critical for localization. To address this, we propose an adaptive graph convolution layer that dynamically learns the direction and degree of message passing, automatically suppressing the propagation of meaningless features while enhancing the propagation of semantic features

$$\mathbf{F}^{\text{out}} = \hat{\mathbf{A}}\mathbf{X}\mathbf{W} \tag{8}$$

where $\mathbf{A}$ denotes the predefined graph adjacency matrix, $\hat{\mathbf{A}}$ denotes the normalized $\mathbf{A}$, $\mathbf{X}$ denotes the input graph node features, and $\mathbf{W}$ is a parameter.

By improving the simple graph convolution layer, the adjacency matrix (node connections) can be learned adaptively based on the input features. Given two local features, a meaningful feature is closer to the global feature than a meaningless feature, which helps to extract uncovered keypoints. Since local node features miss more contextual information, the fusion of local features with global features will enhance contextual associations. Global features refer to extracting depth features from the entire image, and local features refer to the depth features of the target key points. Therefore, this section proposes an adaptive graph convolution reasoning (AGCR) module, which takes the fusion feature $\hat{\mathbf{F}}$ as input, along with a predefined node adjacency matrix. Using the difference between $\mathbf{F}^m$ and $\hat{\mathbf{F}}$, edge weights across all nodes are updated dynamically, leading to $\mathbf{A}^{\text{agcr}}$. Then, a simple graph convolution can be obtained by multiplying $\hat{\mathbf{F}}$ and $\mathbf{A}^{\text{agcr}}$. To stabilize the training process, the input local feature $\mathbf{F}^m$ is fused into the output of the AGCR module. This module is expressed in the following equation, where g_1 and g_2 represent two separate fully-connected layers:

$$\mathbf{F}^S = \left[g_1\left(\mathbf{A}^{\text{agcr}} \otimes \hat{\mathbf{F}}\right) + g_2\left(\hat{\mathbf{F}}\right), \mathbf{F}^m\right]. \tag{9}$$

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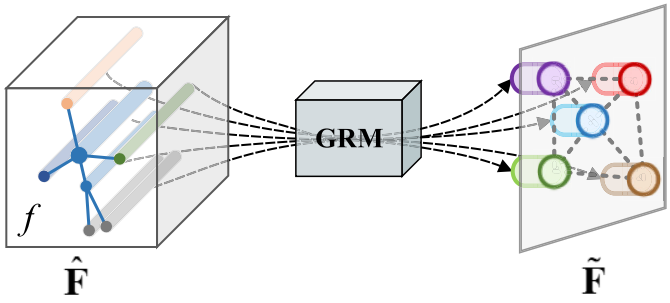


Fig. 6. Schematic of the GRM.

The graph-based reasoning module (GRM) is implemented as a cascade of AGCRs. As shown in Fig. 6, given an image x, with its semantic feature $\mathbf{F}^S = \{f_k^S\}_{k=1}^{K+1}$ obtained by the following equation, its relational feature map $\mathbf{F}^R = \{f_k^R\}_{k=1}^{K+1}$ can be represented as:

$$\mathbf{F}^R = f_{\text{AGCR}}\left(\mathbf{F}^S\right). \tag{10}$$

Meanwhile, the generated parameter $\mathbf{F}^R$ depends on skeleton nodes to dynamically exploit the shape and region information of the current input. This enables motion nodes to rapidly adapt to the scene and fully exploit node relations propagated across history frames for motion graph reasoning. The enhanced feature nodes $\hat{\mathbf{F}}^k$ are obtained by the following equation:

$$\widetilde{\mathbf{F}}^k = f_{\text{GRM}}\left(\mathbf{F}^R, \mathbf{A}\right)^k \tag{11}$$

where $f_{\text{GRM}}(\cdot)$ denotes estimating skeleton nodes using the GRM. The AGCR model adopts the standard keypoint estimation loss function. The training objective is to minimize the total Euclidean or L2 distance between the predicted heatmaps and ground-truth heatmaps across all keypoints. The loss function is defined as follows:

$$L = \frac{1}{K}\sum_{j=1}^{K} v_j \cdot \|T(j) - P(j)\|^2 \tag{12}$$

where $T(j)$, $P(j)$, and v_j denote the ground-truth heatmap, predicted heatmap, and visibility for node j, respectively. The total number of keypoints is set to $K = 9$ during training. The ground-truth heatmap is generated by a 2-D Gaussian centered at the joint location.

E. Antiocclusion Recognition Framework

In this section, the following designs in node matching are introduced. The antiocclusion model employs two different node features to estimate target node locations: one is from the skeleton graph model extracting skeleton nodes and the other is generated by optical flow as motion nodes from previous frames, as shown in Fig. 7. By combining the feature representation of optical flow motion nodes and skeleton shape nodes, this article constructs an antiocclusion recognition model based on a VIDSGN, and the modeling steps are shown in Algorithm 1.

First, for the current frame in the recognition process, the skeleton features of the target are extracted by the skeleton graph model. Target keypoints in the current frame are estimated by target recognition across frames, and then, keypoints across frames are recognized by assigning a unique identification (id) to each of them. The set of target keypoint coordinates is denoted as $J = \{j_i\}_{1:K_n}$. When processing the nth frame image I^n, given the target node set $J^{n-1} = \{J_i^{n-1}\}_{1:K_{n-1}}$ of image I^{n-1} and the target node set $J^n = \{J_i^n\}_{1:K_n}$ of image I^n, where K_{n-1} and K_n are the numbers of nodes in images I^{n-1} and I^n respectively, if the target J_j^n in the current frame I^n is linked to the instance J_i^{n-1} in frame I^{n-1}, then $id_i^{n-1} = id_j^n$; otherwise, a new id is assigned to J_j^n to indicate a new node. Second, the nodes generated by the keypoint detection model and the propagated nodes from previous frames are fused by the optical flow model. The node prediction set generated by propagated nodes is used for complementary predictions and refinement of missing ones from the keypoint detection model, thereby adjusting and predicting target node positions.

Finally, occluded keypoint positions are predicted. The target skeleton graph template is stored in a queue $Q = [J^{n-1}, \ldots, J^{n-L_Q}]$ with a fixed length L_Q. Q can be used to capture linkage relationships from multiple previous frames; it is initialized in the first frame of the image sequence and then updated by the augmenting target set J^n in the nth frame.

Algorithm 1 Antiocclusion Recognition of Aerial IR Target Based on Dual-Stream Graph Network

Input: Video frame $\{I_n\}$, Pipeline of target keypoint sets Q.
Output: keypoint sets J^n, Updated pipeline Q^n.

```
I_0 is input to the backbone network F_key to calculate keypoint locations J^0, and Q = [J^0];
for n = 1:∞ do
    I_n is input to the backbone network F_key to calculate keypoint feature V^c and motion feature V^m;
    Node embedding: F^c = P(V^c), F^m = P(V^m);
    Input to the feature fusion module Ψ(F^m; Θ(F^c)) to calculate fusion feature F̂;
    Input to GRM module f_GRM(·) to calculate enhanced features F̃^k;
    F̃^k is backprojected Γ_{F->V}(·) to get Ṽ, and is fused with the keypoint feature V^c to achieve global prediction J^n.
    Update Q by appending J^n to Q.
```

IV. Experiments

To verify the efficacy and accuracy of the proposed antiocclusion recognition model based on a VIDSGN, this section first introduces the experiment settings and model parameter settings and then validates the efficacy of each module design in the model through a comprehensive ablation study. Finally, recognition experiments are conducted on aerial infrared target datasets using the proposed VIDSGN, with quantitative and qualitative analyses performed on the experimental results.

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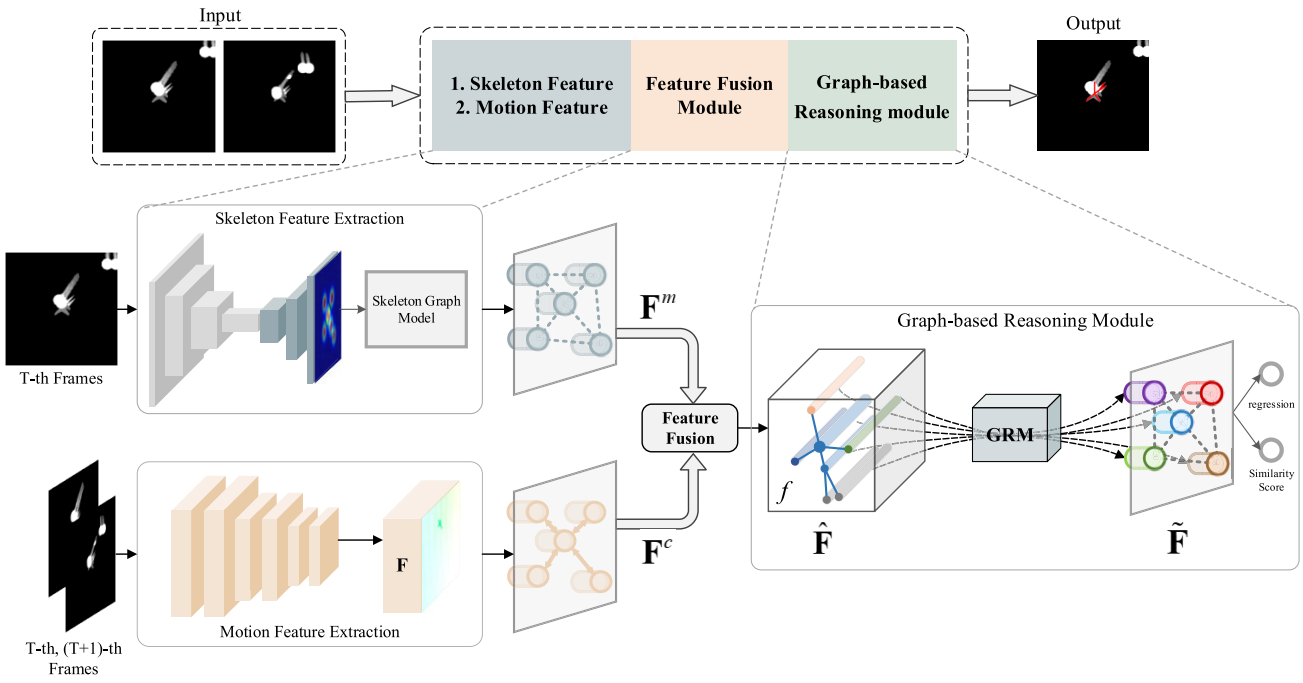


Fig. 7. Antiocclusion recognition framework for air targets based on the VIDSGN.

A. Implementation Details

To evaluate the proposed VIDSGN for handling partial occlusions caused by infrared point interferences, we constructed aerial infrared target datasets using an air combat simulation platform developed in Microsoft[1] Visual Studio. The simulated infrared images used the following parameters: a mid-wave infrared band of 3 $\sim$ 5 μm, a frame rate of 100 Hz, and a resolution of 256 $\times$ 256 pixels. The air combat simulator was built via digital modeling and real-time infrared scene rendering, comprising models of the target, infrared interference, and background.

1) Fluent[2] software was utilized for flow field computation, with model calibration performed using field measurements.
2) The open scene graph (OSG) rendering backend and OpenGL parallel rendering were adopted for real-time infrared scene generation and dataset output to synthesize infrared imagery. MODTRAN was used to select the atmospheric mode and configure related parameters. Finally, the infrared target model was imported into the GPU for parallel computing.
3) The simulation of the confrontation process between the target and aircraft involves confrontation situation design and parameter quantification under different conditions and scenes for the simulation system, mainly including the aircraft motion model, interference infrared simulation model, interference motion model, and so on. As listed in Table I, the generated simulation dataset consists of 72 sequences, covering various situations such as the target's large maneuver turning, target posture rotation, and target maneuvering accompanied by infrared interference. The setting of the ballistic parameters when generating the dataset includes the initial entry angle selected at every 10° from 0° to 360° and the infrared decoy launching strategy: 24 groups in total, with 2 or 4 decoys per group, and 0.7 s between groups. The simulation dataset contains 86 746 frames in total. Since this article focuses on the model's efficacy in target recognition under occlusion, the ratio of frames with interference occlusion over the total number of frames is statistically 81.2% in the simulation dataset.

TABLE I
DATASET COMPOSITION TYPE AND THE RELATED PARAMETERS USED IN THE EXPERIMENT

dataset type	Simulation	Measurement
Infrared band	$3 \sim 5\mu m$	$3 \sim 5\mu m$, $7.5 \sim 14\mu m$
resolution	256×256	640×512
Sequence number	72	9
Number of frames	86,748	7,858
Ratio of frames occluded	81.2%	60.1%
Type of interference	infrared decoys	background clutter, infrared detector noise, infrared decoys
Delivery strategy of interference	24 groups, with 2 or 4 decoys per group	-
Maneuver type	left maneuver, right maneuver, dive, climb	roller and scale variation, maneuver variation

Meanwhile, to validate the proposed VIDSGN in dealing with challenges of background interference, infrared decoy interference, target maneuvering, and so on under real scenarios, this article constructs an aerial infrared real-world dataset using flight target data collected by an infrared thermal imager. The dataset consists of nine sequences with the following

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specifics: the infrared bands include mid-wave 3∼5 μm and long wave 7.5∼14 μm, the frequency is 100 Hz, and the resolution is 640 × 512 pixels. This dataset is shot on the stationary outdoor ground of various aerial targets during flight, containing complex cloud background interference, detector noise interference, target maneuver turning, posture rotation changes, and so on. Statistically, the real-world dataset contains 7858 frames in total, where the ratio of frames with interference occlusion to the total number of frames is 60.1%, including infrared point decoy occlusion and cloud cluster background interference. To sum up, the aerial infrared target dataset constructed in this article contains simulation and real-world data; 80% of the simulation sequences and 80% of real-world sequences are selected as the training set, and the remaining 20% sequences are used for testing, as shown in Table I.

In the model training process, the number of training epochs is 120. The entire framework is trained with each batch containing 32 images. The initial learning rate is set to $1e-3$ and decays in a cosine manner. The Adam optimizer is used for the backpropagation optimization of parameters. This model is implemented in PyTorch [49], and experiments are conducted on a Nvidia[1] GeForce GTX[2] 3090 graphics card with 24-GB memory. For comparison, state-of-the-art general recognition models are tested, including HRNet(W48) [50], HigherHRNet(W48) [51], AE [52], OpenPose [53], ResNet [54], ResNext [55], Mask-RCNN [39], AlphaPose+ [56], SimplePose [36], Graph-PCNN [57], MF-GCN [2], PoseWarper [58], and MSGSM [15]. For a fair comparison, the public code of Mask-RCNN and SimplePose from [56] is run directly, and AlphaPose+ is retrained on the dataset designed in this article.

Based on point annotations and test results, evaluation metrics are proposed in this article based on similarity. When evaluating comparison models on the test set, this article focuses on the object keypoint similarity (OKS) of the target

$$\mathrm{OKS} = \frac{\sum_i \exp\left(-d_i^2/2s^2t_i^2\right) \cdot \delta(v_i > 0)}{\sum_i \delta(v_i > 0)} \tag{13}$$

where d_i denotes the Euclidean distance between the ground-truth point and the predicted point, and v_i denotes the visibility flag of the standard point.

Following common practice in existing work, we adopt the standard evaluation metrics from MSCOCO [59] based on keypoint similarity to assess aerial target keypoint prediction accuracy. Based on the keypoint similarity results, we report average precision (AP) and average recall (AR) to evaluate the models, using IoU thresholds of 0.5:0.95, 0.5, and 0.75 for AP. Meanwhile, two criteria were defined to select valid samples meeting the training requirements of the proposed VIDSGN: the target keypoint prediction must contain at least three visible points with an OKS score above 0.25 to ensure the quality of training.

B. Ablation Study

To thoroughly analyze the proposed VIDSGN, comprehensive ablation experiments are conducted to evaluate the capability of each module in VIDSGN. As shown in Table II, baselines are provided to investigate the impact of each module. For clarity, the graph-based skeleton module guided by keypoints is denoted as GSM, the node optical flow module guided by optical flow features is denoted as NOF, and the GCN-based reasoning module guided by fused features is denoted as GRM. To demonstrate the efficacy of our model and the contribution of each module, experiments are first conducted by adding one module at a time in this section. Then, performance experiments are conducted on antiocclusion pose estimation for aerial targets on the dataset. The advantage of VIDSGN over other methods is proven in terms of mAP.

TABLE II
ABLATION STUDY OF THE ANTIOCCLUSION RECOGNITION MODEL

Models	Baseline	GSM	NOF	GRM	Stage1 mAP	Stage2 mAP
VIDSGN_V1	✓	×	×	×	0.672	-
VIDSGN_V2	✓	✓	×	×	0.694	-
VIDSGN_V3	✓	✓	✓	×	0.743	0.791
VIDSGN_V4	✓	✓	✓	✓	0.786	0.827

This article further evaluates the efficacy of the proposed antiocclusion recognition framework. As presented in Table II, compared to the keypoint detection in the first stage, the second stage improves mAP by 18.7% and 16.5%, respectively, indicating that the antiocclusion recognition framework based on regression is more effective than direct heatmap regression of keypoint positions. Meanwhile, jointly training the overall framework improves the performance of the first stage on heatmap estimation of keypoints by 2.2%, 7.1%, and 11.4% in mAP compared with the first-stage model alone.

This study further investigates each design of the proposed VIDSGN. From VIDSGN-V1 and V2 to V4, the progressive coarse-to-fine feature learning strategy is validated to be effective. Meanwhile, the proposed cascade modules are analyzed. In Table II, when the GSM and NOF modules are removed, the mAP on the dataset drops significantly, demonstrating the indispensable role of the modules in the overall framework. Then, the GRM module is removed, and the results are reported in Table II, which further demonstrates the efficacy of the reasoning part in the GRM module. Thus, it can be concluded that image guidance is the most crucial in the framework. The average mAP gain of the GRM module on the dataset is 4.3, indicating the necessity to adapt image features for the reasoning module. Overall, these ablation studies demonstrate the efficacy of each component of the proposed model, with the various cues providing complementary information for invisible joint inference.

C. Qualitative Analysis

This section performs a qualitative analysis of the experimental results between VIDSGN and comparison models. As shown in Figs. 8 and 9, VIDSGN can correct wrong connections between nodes and estimate occluded nodes while achieving accurate recognition of visible nodes. The following observations can be made from the results: 1) for sequences with target scale variations, posture rotation variations, and fast maneuvering variations, VIDSGN achieves good target

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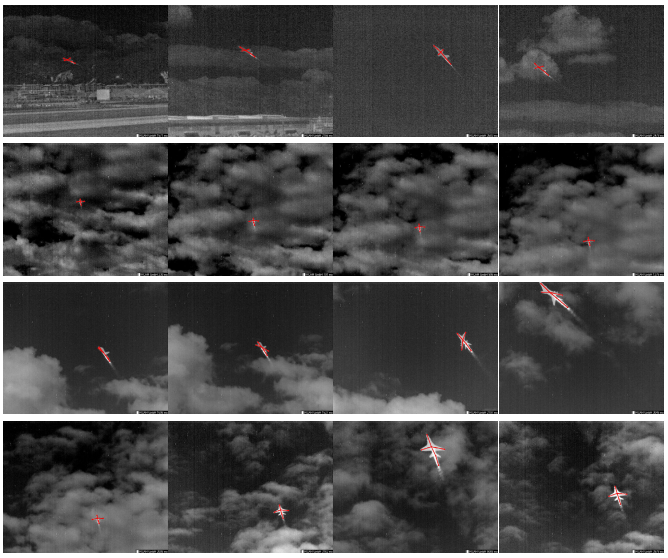


Fig. 8. Recognition results of measured infrared datasets.

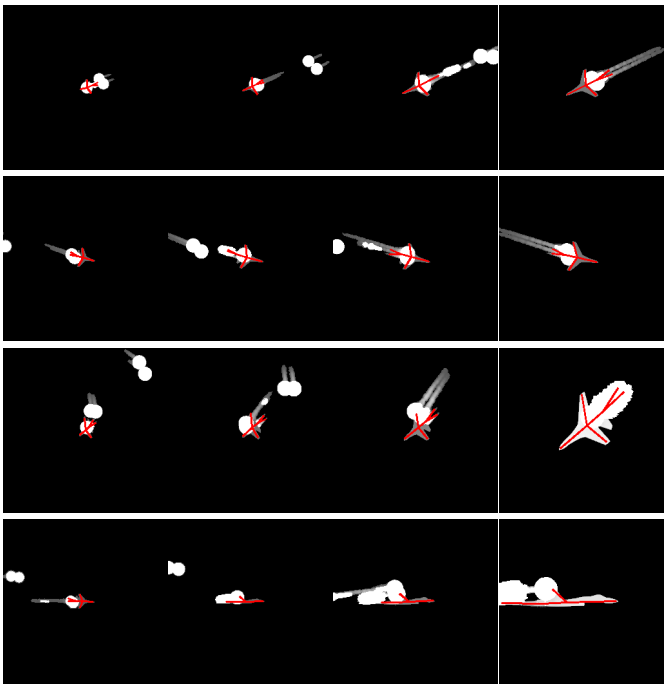


Fig. 9. Antiocclusion recognition results on the simulated infrared datasets.

recognition results; even relatively large displacements, scale changes, and fast barrel rolls can be corrected by the skeleton graph module and node optical-flow module; 2) for situations with ground clutter background and cloud background interference, VIDSGN can still adjust the joints to the correct positions; and 3) in the presence of infrared decoys, VIDSGN produces a more natural target skeleton structure. Even with severe target occlusion due to interference, VIDSGN can still infer plausible target node positions that adhere to the physical constraints between key target regions. This analysis indicates that VIDSGN is capable of estimating correct connections between occluded target skeleton nodes.

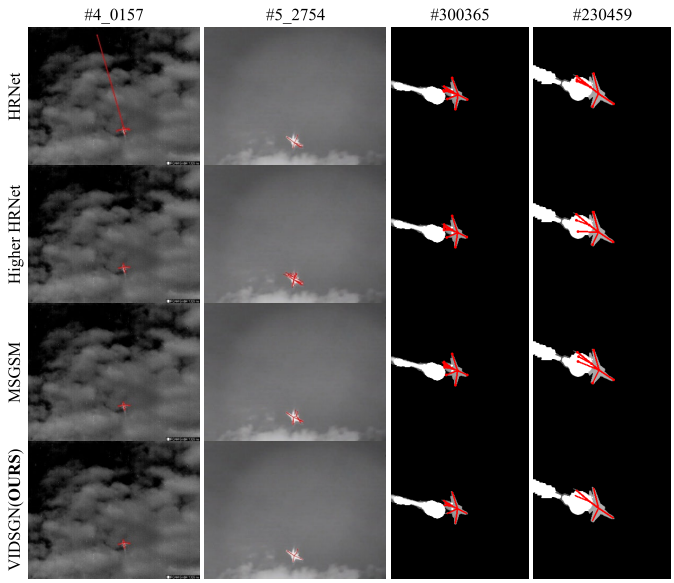


Fig. 10. Side-by-side comparison of model recognition results.

For qualitative analysis of test results to evaluate the capability of VIDSGN in adapting to complex scenes, side-by-side comparisons to state-of-the-art methods are illustrated in Fig. 10. Each column displays a different scene, including fast motion, background interference, occlusion interference, and so on, and each row shows the recognition results from a different method. It can be observed from the figure that the model proposed in this article demonstrates more robust recognition effects across various scenes. ResNet [54] and HRNet [50] are trained on static images, so they cannot capture temporal dependency across video frames, leading to suboptimal recognition results. Meanwhile, MSGSM [15] utilizes temporal information between video frames to warp the initial heatmaps, and it uses an adjacent frame as an auxiliary. The model proposed in this article makes full use of temporal context when processing consecutive frames. By designing the NOF module to better encode motion information and the SGM module to better encode target skeleton features, the recognition accuracy of the target skeleton module is further enhanced. Thus, VIDSGN achieves high performance both quantitatively and qualitatively.

D. Quantitative Analysis

For quantitative analysis of test results, the proposed evaluation metrics and dataset are adopted to evaluate the test results of VIDSGN and comparison models. Table III presents the quantitative results of different methods in terms of the mAP metric. Compared with five existing recognition models, VIDSGN achieves an average recognition rate of 85.9% on the multiframe aerial target recognition task on the test set. It improves the mAP by 12.1% compared to MSGSM and 7.5% compared to the multiframe recognition model MF-GCN. Besides, VIDSGN outperforms all comparison models comprehensively on evaluation metrics.

The above qualitative and quantitative analyses indicate that the model proposed in this article achieves improved

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TABLE III
COMPARISON OF ANTIOCCLUSION RECOGNITION RESULTS BASED ON KEY POINT ESTIMATION

Model	Backbone	mAP	AP^{50}	AP^{75}	AP^{M}	AP^{L}	AR
AE [52]	—	0.615	0.808	0.663	0.556	0.696	0.672
OpenPose [53]	VGG19	0.618	0.769	0.615	0.581	0.682	0.675
HRNet [50]	HR48	0.704	0.851	0.692	0.661	0.757	0.746
HigherHRNet [51]	HR48	0.715	0.847	0.713	0.669	0.765	0.755
ResNet [54]	Res101	0.618	0.819	0.675	0.571	0.682	0.665
ResNext [55]	Res101	0.649	0.855	0.713	0.623	0.700	0.697
Mask-RCNN [39]	R50-FPN	0.631	0.833	0.687	0.578	0.714	—
AlphaPose+ [56]	Res101	0.698	0.862	0.741	0.695	0.802	0.732
SimplePose [36]	Res101	0.631	0.865	0.712	0.612	0.685	0.698
Graph-PCNN [57]	HR48	0.707	0.854	0.716	0.610	0.687	0.703
MSGSM [15]	HR48	0.738	0.874	0.752	0.704	0.817	0.723
PoseWarper [58]	HR48	0.649	0.855	0.713	0.623	0.700	0.697
MF-GCN [2]	HR48	0.783	0.902	0.813	0.749	0.876	0.751
VIDSGN(Ours)	HR48	**0.859**	**0.936**	**0.854**	**0.783**	**0.927**	**0.793**

performance in target recognition, and it can predict target keypoints even with severe partial occlusion, obtaining a target recognition rate of 85.9%. According to the analysis of the test results, the reason why the proposed model succeeds in recognizing heavily occluded targets on the constructed infrared dataset is mainly that it introduces target motion feature extraction and fusion strategies, which can still accurately predict occluded node positions under occlusion. This benefits from the proposed feature fusion module and graph reasoning module GRM, which combines the advantages of skeleton shape features and motion features to better exploit spatiotemporal characteristics of keypoints, thereby estimating occluded node positions, better predicting target keypoints, and improving antiocclusion recognition performance.

After further analyzing the failed recognition data (14.1%) from the motion feature dimension and locating the problem, some failed recognition results of VIDSGN are shown in Fig. 11. On the one hand, in very few cases in the mid-to-near stage, there are situations where only 1∼2 keypoints are unoccluded. The graph reasoning module designed in this article requires features of multiple associated skeleton nodes and significant motion features to improve the accuracy of occluded node estimation. Severe occlusion also easily leads to insignificant target motion features, making it more difficult to predict the positions of occluded target keypoints. On the other hand, most of the recognition failures are in the far stage, where the target is fully occluded, and motion features and skeleton features are completely missing. The antiocclusion recognition model proposed in this article cannot estimate keypoint positions by extracting motion and skeleton features, resulting in model recognition failures easily. To sum up, the main unresolved issue of VIDSGN is the inability to extract target visual features under full occlusion.

E. Antiocclusion Challenge

In order to further verify the anti-interference recognition performance of VIDSGN in this article during the full interference occlusion stage, part of the image sequence of the target interference is intercepted from the infrared dataset, and the antiocclusion performance of the model is tested and



Fig. 11. Failure of antiocclusion.

TABLE IV
COMPARISON OF ANTIOCCLUSION RECOGNITION RESULTS BASED ON KEY POINT ESTIMATION

Model	Backbone	mAP	AR
AlphaPose+ [56]	Res101	0.283	0.351
SimplePose [36]	Res101	0.267	0.292
Graph-PCNN [57]	HR48	0.369	0.316
MSGSM [15]	HR48	0.394	0.347
PoseWarper [58]	HR48	0.336	0.277
MF-GCN [2]	HR48	0.417	0.384
VIDSGN(Ours)	HR48	**0.593**	**0.567**

analyzed in a targeted manner. Since the severe occlusion situations studied in this article mainly occur in the medium- and long-range stages, and the interference affects the target subject more seriously, the image sequences intercepted in this experiment are concentrated in the medium- and long-range stages. After filtering and intercepting the infrared simulation dataset, the image sequence with severe interference and occlusion of the target constitutes the occlusion dataset, which contains 51 fragments and 5269 images.

In the occlusion dataset, our proposed VIDSGN and comparison models were further tested for target recognition performance under severe occlusion. Table IV shows the quantitative analysis results of antiocclusion recognition for all algorithms. It can be seen that the target recognition model based on the graph model proposed in this article has better recognition results in severe occlusion situations. In terms of recognition accuracy, the VIDSGN proposed in this article is 17.6% higher than MF-GCN, 19.8% higher than MSGSM, and 25.7% higher than PoseWarper. The visual features of the target almost completely disappear under severe occlusion. The VIDSGN proposed in this article can estimate the semantic information of the target and interference based on the spatiotemporal correlation information of the target and interference and then estimate the target position. Other comparison algorithms estimate the target position through visual features. In the case of occlusion, the target features disappear and the target position cannot be estimated.

V. CONCLUSION

To address the issue of partial occlusion during aerial target recognition caused by infrared point interferences, this article proposes an antiocclusion recognition model based on a VIDSGN to alleviate target identification loss caused by brief full occlusion of aerial targets. First, the model learns target skeleton features and motion features through the graph-based skeleton module and the node optical-flow module, respectively, and it aggregates the two types of feature representations in the reasoning grid to enhance the capability

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of occluded node prediction. Second, this article designs an adaptive parameterized graph convolution operation, which can dynamically adjust message passing direction and strength based on different inputs to suppress meaningless feature propagation. Besides, cascade multisemantic feature extraction modules are designed to achieve coarse-to-fine feature expression. Finally, model training, testing, and performance analysis are carried out by constructing an infrared target dataset containing both simulation and real-world data. Experimental results show that the average recognition accuracy of VIDSGN reaches 85.9%, and it enhances keypoint prediction capability under occlusion with certain robustness in complex environments. However, full occlusion where target features cannot be extracted is still a limitation of the proposed model.

However, further improvements are needed to overcome the challenge of target recognition under full occlusion. In particular, the situation of interference with long-term full occlusion accompanied by rapid target maneuvers remains unresolved. This is also an engineering problem encountered in the current development of anti-interference recognition technology. We plan to use limited visual input signals to judge and predict the flight position of the target during long-term total occlusion in future work. This is a direction that requires research and challenges for a long time to come.

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